

Filename: EAPS120-HW2

Earth, Atmospheric, and Planetary Sciences 120 - Homework Assignment 2

Student Name: _____ | Due Date: Fall 2024

Weather and Climate Problem Set

Problem 1

Calculate the dry adiabatic lapse rate. If a parcel of dry air rises from sea level ($T = 20^{\circ}\text{C}$, $P = 1013.25 \text{ hPa}$) to 2000 m elevation, what will be its temperature? Assume no mixing with surrounding air.

Solution:

The dry adiabatic lapse rate (Γ_d) is given by:

$$\Gamma_d = g/c_p$$

where $g = 9.81 \text{ m/s}^2$ (acceleration due to gravity) and $c_p = 1004 \text{ J/(kg}\cdot\text{K)}$ (specific heat at constant pressure for dry air).

$$\Gamma_d = 9.81 / 1004 = 0.0098 \text{ K/m} = 9.8 \text{ K/km}$$

For a parcel rising from sea level to 2000 m:

$$\Delta z = 2000 \text{ m}$$

$$\Delta T = -\Gamma_d \times \Delta z = -9.8 \text{ K/km} \times 2 \text{ km} = -19.6 \text{ K}$$

$$\text{Final temperature: } T = 20^{\circ}\text{C} - 19.6^{\circ}\text{C} = 0.4^{\circ}\text{C}$$

Note: The negative sign indicates cooling as the parcel rises and expands adiabatically.

Problem 2

A weather balloon measures the following temperature profile:

- Surface: 15°C
- 1000 m: 10°C
- 2000 m: 8°C
- 3000 m: 5°C

- Calculate the environmental lapse rate for each 1000 m layer.
- Determine the atmospheric stability for a surface parcel at each level.
- Identify any temperature inversions.

Solution:

a) Environmental lapse rate (Γ) = $-\Delta T/\Delta z$

- 0-1000 m: $\Gamma = -(10-15)/(1000-0) = 5 \text{ K/km}$
- 1000-2000 m: $\Gamma = -(8-10)/(2000-1000) = 2 \text{ K/km}$
- 2000-3000 m: $\Gamma = -(5-8)/(3000-2000) = 3 \text{ K/km}$

b) Atmospheric stability depends on comparing environmental lapse rate (Γ) with dry adiabatic lapse rate ($\Gamma_d = 9.8$ K/km):

- If $\Gamma > \Gamma_d$: Unstable (parcel will continue rising)
- If $\Gamma = \Gamma_d$: Neutral
- If $\Gamma < \Gamma_d$: Stable (parcel will sink back)

For a surface parcel ($T = 15^\circ\text{C}$):

- At 1000 m: Parcel temp = $15 - 9.8 = 5.2^\circ\text{C}$, Environment = 10°C . Parcel cooler, so STABLE.
- At 2000 m: Parcel temp = $15 - 19.6 = -4.6^\circ\text{C}$, Environment = 8°C . Parcel much cooler, STABLE.
- At 3000 m: Parcel temp = $15 - 29.4 = -14.4^\circ\text{C}$, Environment = 5°C . Parcel cooler, STABLE.

All layers are stable because $\Gamma < \Gamma_d$ in each case.

c) No temperature inversions are present. An inversion would require temperature increasing with height, but all temperatures decrease with height in this profile.

Problem 3

The relative humidity is 60% at a temperature of 25°C . The air pressure is 1000 hPa.

- Calculate the actual vapor pressure.
- Find the dew point temperature.
- If the temperature drops to 15°C , what will be the relative humidity?

Solution:

a) First, find the saturation vapor pressure at 25°C using the Clausius-Clapeyron equation or approximate formula:

$$e_s(25^\circ\text{C}) \approx 6.11 \times 10^{\frac{7.5T}{237.7+T}} \text{ hPa}$$

$$e_s(25^\circ\text{C}) \approx 6.11 \times 10^{\frac{7.5 \times 25}{237.7+25}} = 31.7 \text{ hPa}$$

$$\text{Relative humidity RH} = (e/e_s) \times 100\%$$

$$0.60 = e / 31.7$$

$$e = 0.60 \times 31.7 = 19.0 \text{ hPa (actual vapor pressure)}$$

b) Dew point is the temperature at which the air becomes saturated ($e_s = e = 19.0$ hPa).

$$\text{Using the inverse relationship: } T_d = \frac{237.7 \times \log_{10}(e/6.11)}{7.5 - \log_{10}(e/6.11)}$$

$$T_d = \frac{237.7 \times \log_{10}(19.0/6.11)}{7.5 - \log_{10}(19.0/6.11)}$$

$$T_d \approx 16.7^\circ\text{C}$$

c) At $T = 15^\circ\text{C}$, the saturation vapor pressure is:

$$e_s(15^\circ\text{C}) \approx 6.11 \times 10^{\frac{7.5 \times 15}{237.7+15}} = 17.0 \text{ hPa}$$

The actual vapor pressure remains $e = 19.0$ hPa (assuming no condensation occurs).

$$\text{RH} = (19.0 / 17.0) \times 100\% = 112\%$$

Since $\text{RH} > 100\%$, the air is supersaturated and condensation will occur. The relative humidity cannot exceed 100% in equilibrium, so excess water vapor will condense, maintaining $\text{RH} = 100\%$.

Problem 4

Explain the difference between weather and climate. Give three examples of climate change indicators and explain how each relates to Earth's energy balance.

Solution:

Weather vs. Climate:

Weather refers to short-term atmospheric conditions (hours to days) at a specific location, including temperature, precipitation, wind, and humidity. Climate refers to long-term average weather patterns (typically 30+ years) and variability for a region.

Climate Change Indicators:

1. **Global Average Temperature:** Rising global temperatures indicate an energy imbalance where more energy is being absorbed than radiated. This is primarily due to increased greenhouse gas concentrations trapping outgoing infrared radiation.
2. **Sea Level Rise:** Rising sea levels result from both thermal expansion of ocean water (as it warms) and melting of land-based ice. This reflects increased energy absorption by the climate system, particularly in the oceans which store over 90% of the excess heat.
3. **Arctic Sea Ice Extent:** Declining Arctic sea ice reduces Earth's albedo (reflectivity), creating a positive feedback loop. Less ice means more solar radiation is absorbed by dark ocean water instead of being reflected, further warming the system.

All three indicators are interconnected through Earth's energy balance equation: $\text{Energy In} = \text{Energy Out}$. When greenhouse gases increase, less infrared radiation escapes to space, causing the system to warm until a new equilibrium is reached.